

Experiment (1):

To measure diameter of a small spherical body, using vernier callipers.

APPARATUS

Vernier callipers, A spherical body (pendulum bob).

THEORY

If with the body between the jaws, the zero of vernier scale lies ahead of N th division of mainscale, then main scale reading (M.S.R.) = N .

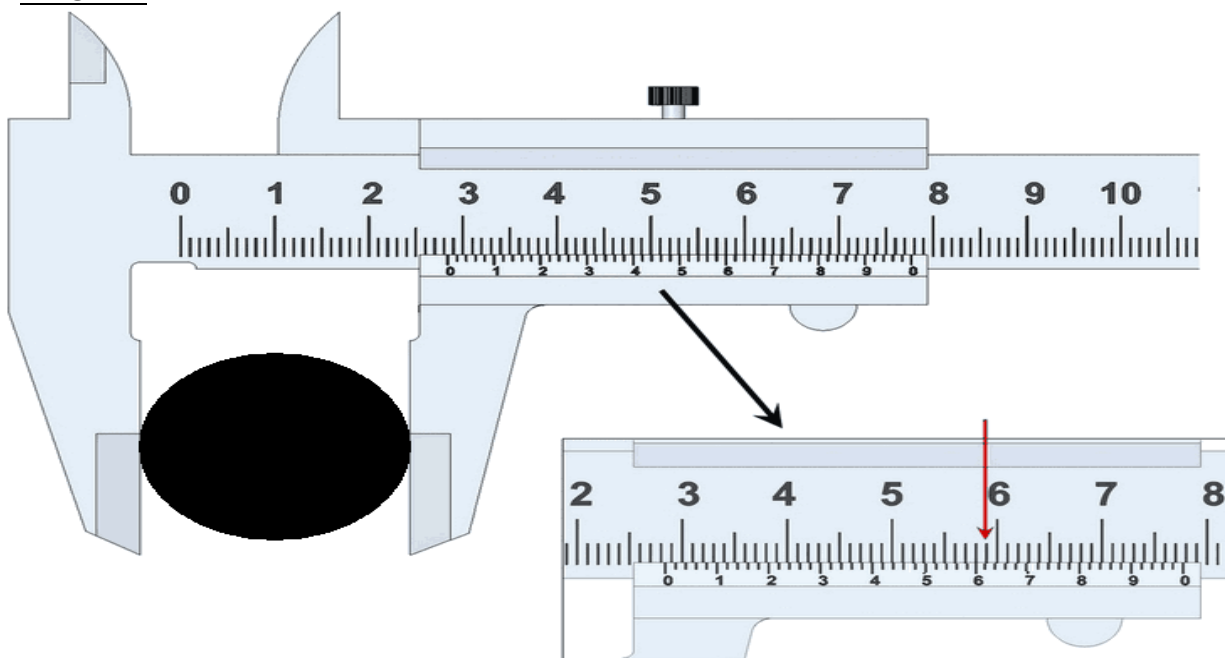
If n th division of vernier scale coincides with any division of main scale, then Vernier scale reading (V.S.R.)

= $n \times (\text{L.C.})$ where L.C. is least count of vernier callipers.

= $n \times (\text{V.C.})$ where V.C. is vernier constant of vernier callipers

Total reading, T.R. = M.S.R. + V.S.R. = $N + n \times (\text{V.C.})$.

DIAGRAM



Vernier calipers measuring diameter of a sphere.

PROCEDURE

1. Determine the vernier constant (V.C.) i.e. least count (L.C.) of the vernier callipers and record it stepwise.
2. Bring the movable jaw BD in close contact with the fixed-jaw AC and find the zero error. Do it three times and record them. If there is no zero error, then record 'zero error nil'.
3. Open the jaws, place the sphere or cylinder between the two jaws A and B and adjust the jaw DB, such that it gently grips the body without any undue pressure on it. Tight the screw S attached to the vernier scale V.
4. Note the position of the zero mark of the vernier scale on the main scale. Record the main scale reading just before the zero mark of the vernier scale. This reading (N) is called main scale reading (M.S.R.).

- Note the number (n) of the vernier scale division which coincides with some division of the main scale.
- Repeat steps 4 and 5 after rotating the body by 90° for measuring the diameter in perpendicular direction.
- Repeat steps 3, 4, 5 and 6 for three different positions. Record the observations in each set in a tabular form.
- Find total reading and apply zero correction.
- Take mean of different values of diameter and show that in the result with proper unit

Observations:

Vernier Constant (Least Count) of the Vernier Callipers:

1 M.S.D. = 1 mm = 0.1 cm

No. of V.S.D (N) = 50

$$\begin{aligned}\text{Vernier constant V.C} &= (1/N) \times 1 \text{ MSD} \\ &= (1/50) \times 0.1 \text{ cm} \\ &= 0.002 \text{ cm}\end{aligned}$$

I.E. (X) = $\pm x \times \text{V.C.}$ (Where x = The VSD that coincide with any MSD with the jaws closed).

Table: Slide calipers used to determination of diameter of a sphere/cylinder:

No. of obs.	M.S.R. (A) cm	V.S.D. (B)	Value of VSR C = B $\times$ V.C. cm	Total $D_1 = A + C$ cm	I.E. $\pm X$ cm	Corrected reading D = $D_1 - (\pm X)$ cm	Mean D cm.
1						$X_0 = \dots\dots$	$(X_0 + X_1 + X_2)/3$
2						$X_1 = \dots\dots$	
3						$X_2 = \dots\dots$	

Calculations:

Mean diameter **D** = cm

RESULT:

The diameter of the given sphere/cylinder is **D** = cm.

PRECAUTIONS

- Motion of vernier scale on main scale should be made smooth.
- Vernier constant and zero error should be carefully found and properly recorded.
- The body should be gripped between the jaws firmly but gently.
- Observations should be taken at right angles at one place and taken at least at three different places.

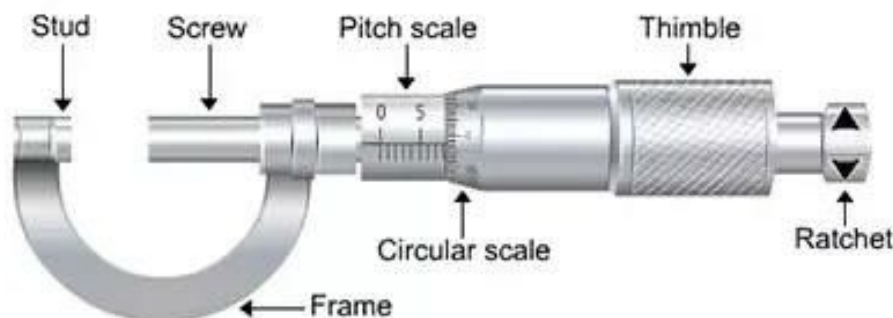
EXPERIMENT: DETERMINE THE DIAMETER OF A GIVEN WIRE USING A SCREW GAUGE.

Apparatus used:

Given wire, Screw gauge.

THEORY:

Screw Gauge is a mechanical tool that facilitates measuring the diameter or radius or thickness of a thin wire or thickness of a thin metal sheet with utmost accuracy. Figure 1 shows the schematic of the Screw Gauge.



(i) **Pitch:** The pitch of the screw is the distance through which the screw moves along the main scale in one complete rotation of the cap on which is engraved the circular scale.

(ii) **Least Count:** The least count of the screw gauge is the distance through which the screw moves when the cap is rotated through one division on the circular scale.

(iii) **Zero error and correction:** When the opposite studs on the frame are brought into contact, if the zero of the head scale coincides with the circular scale axis, there is no zero error (See Fig. 1(a)).

When the zero mark of the circular scale and the main scale do not coincide on bringing the studs in contact the instrument has zero error. The zero of the circular scale may be in advance or behind the zero of the main scale by a certain number of divisions on circular scale.

If the zero of the circular scale is behind the zero of the pitch scale, the zero error is positive (See Fig. 1 (b)).

On the other hand, If the zero of the circular scale is ahead of the zero of main scale the zero error is negative (See Fig. 1 (c)).

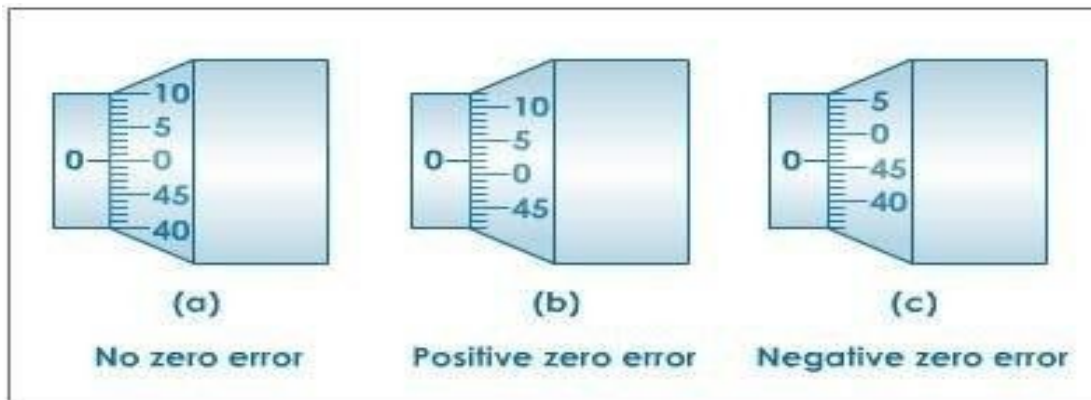


Fig.1

(iv) **Back-lash error:** Owing to ill-fitting or wear between the screws and the nut, there is generally some space for the play of the screw, the screw may not move along its axis for appreciable rotation of the head (or cap, on which circular scale is marked). The error so introduced is called back-lash errors. To eliminate it you must advance the screw, holding it by the retched cap, when making final adjustment for finding zero error or the diameter of the wire.

Procedure:

(i) **Measuring pitch:** To measure the pitch give several rotation to its cap and observe the distance through which screw moves. Calculate the pitch using the following formula.

$$\text{Pitch} = (\text{Distance moved}) / (\text{No of complete rotation})$$

(ii) **Measuring least count:** To measure the least count note the number of divisions on the circular scale and calculate

$$\text{Least Count (L. C)} = (\text{Pitch of the screw gauge}) / (\text{No. of division on circular scale})$$

(iii) **Measuring zero error:** With the studs in contact observe the numbers of divisions by which zero of the circular scale deviates from the zero of the main scale. This number multiplied by the least count gives the required zero error (As discussed previously).

(iv) **Calculate the zero correction:** It is negative of zero error.

Zero correction = – zero error (Where zero error is put with its proper sign).

Zero correction is added algebraically in the observed diameter of wire to get the corrected reading.

(v) **Measuring diameter:** To measure the diameter of the wire moves the screw back to make a gap between the studs. Insert the wire between the studs. Turn the screw forward by holding it from the ratchet cap and wire should be held gently between the two studs.

(vi) Read the nearest division on the circular scale in line with the main scale and also find the complete rotations of the cap with the help of the main scale. Calculate the observed diameter:

Observed diameter = Pitch $\times$ number of complete rotation + L.C. $\times$ circular scale reading

- (vii) Repeat the experiment for 5 observations at different points of the wire along its length. Find the mean observed diameter and apply the zero correction to obtain correct diameter.

Observations:

Pitch of screw (P) =cm. (Linear scale distance covered in 1 complete rotation of circular scale)

Total no. of circular scale div. (C) =

L.C. = P/C =cm.

Instrumental Error (I.E.) $X = \pm (X \times L.C.)$ cm

(Where X = The CSD that coincide with the reference line with the jaws closed)

Table1: Reading for diameter of the wire:

No. of obs.	L.S. Reading (A) cm	C.S.D. (B)	C.S.R. $C = (B \times L.C.)$ cm	Total $D_1 = A + C$ cm	I.E. $\pm X$ cm	Corrected Diameter $D = (d_1 \pm X)$ cm.	Mean D cm
1						$X_0 = \dots\dots\dots$	$(X_0 + X_1 + X_2)/3$
2						$X_1 = \dots\dots\dots$	
3						$X_2 = \dots\dots\dots$	

Mean corrected diameter (D) = cm

Precautions:

1. There shouldn't be any friction while moving the screw.
2. The screw should always be rotated by ratchet R and not by cap K to avoid undue pressure.
3. For zero correction proper sign should be noted.
4. To avoid back-lash error in the screw, the screw should be moved in the same direction.
5. The diameter of the wire should be measured at each place in two perpendicular directions and mean should be taken.
6. Along the entire length of the wire, readings should be taken at five different places

Experiment (2):

To use a Multimeter for measuring

- a) Resistance
- b) Capacitance
- c) Inductance
- d) AC and DC Voltages
- e) DC Current,

Apparatus used: Multimeter, Connecting wires, Resistance, DC battery/power supply, Fuse.

Theory:

The Basics of Digital Multimeters :

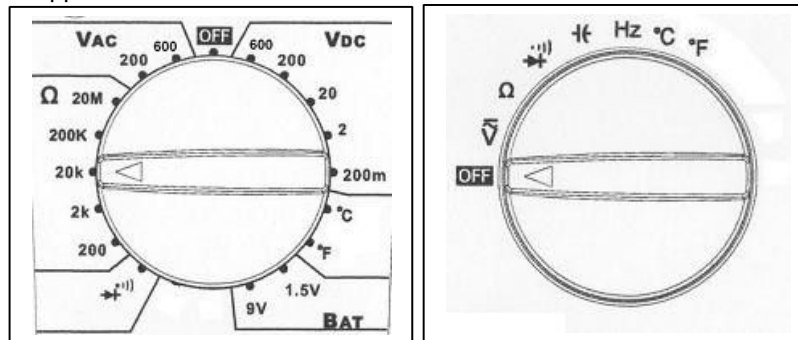
There are two common types of Multimeters, Analog and Digital. Digital Multimeters (DMMs) are the most common.

Analog meters use a needle movement and calibrated scale to indicate values. These were popular for years, but recently their numbers have declined

The Digital Multimeter (DMMs) feature a digital or liquid crystal display (LCD). Measurement readings are displayed as numerical values on the LCD Display.

The Dial Setting the Function The dial of the DMM allows you to choose the function you're interested in measuring. Whether you intend to measure the resistance, frequency or capacitance, you must first set the dial to the appropriate function.

Setting the Range: The dial also plays another essential role in measuring electricity – that of determining the range of measurement. The range you select on the dial determines the placement of the decimal point as it appears on the LCD.



On a manual ranging meter, the function and range must be selected / Auto ranging meters will automatically choose the measurement range

Symbols	Measurement Functions	Descriptions
	AC Voltage	Measures amount of AC Electrical Pressure
	DC Voltage	Measures amount of DC Electrical Pressure
mV	Milli Volts	.00V or 1/1000V
A	Amperes	Measures amount of electron flow
mA	Milli Amperes	.001 or 1/1000A
	Ohms	Measurement of resistance to the flow of electron
	Diode	Device used to control direction of electron flow
	Audible Continuity	Audible indication of continuity for low resistance

	Capacitance	Device used to store electrical potential

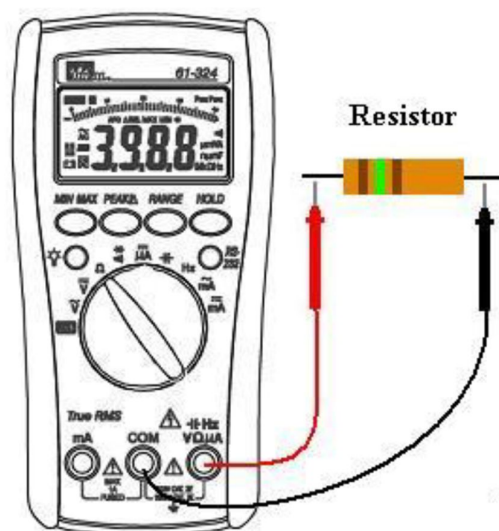
The input jacks or ports of your meter are the working ends of the instrument. Use care when connecting leads to your instrument. Pay close attention and be sure to connect the leads into the correct port that is marked for the measurement selected on the dial.



PROCEDURE:

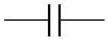
Resistance Measurements

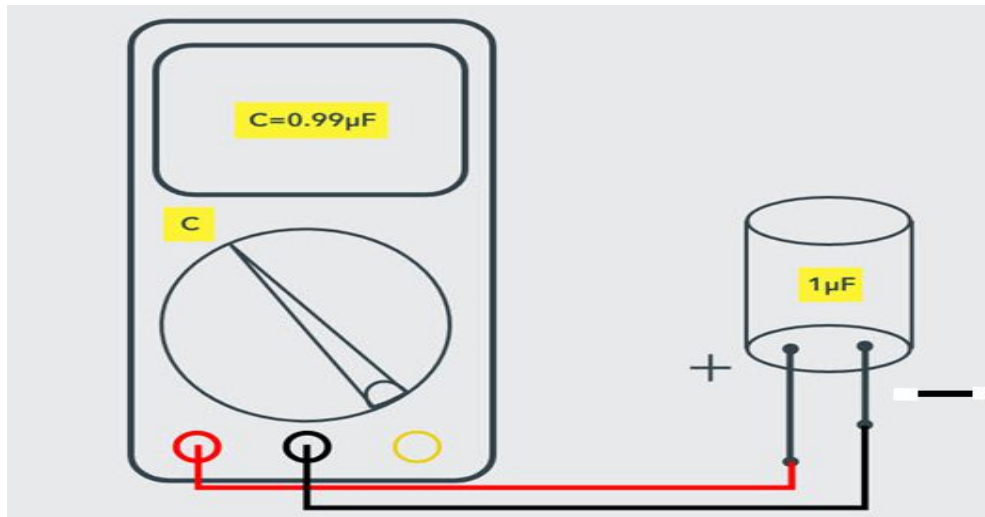
Resistance measured in ohms (Ω). It is important when measuring Resistance that the circuit be de-energized or turned off, or the circuit may damage the meter. Most meters have overload protection on all ranges to prevent this, but you should check the specifications of your digital multimeter to be sure.



For resistance measurements, place the test leads on each side of the resistor. Other components in parallel with the resistor being measured will have an effect on the measurement.

Capacitance Measurements

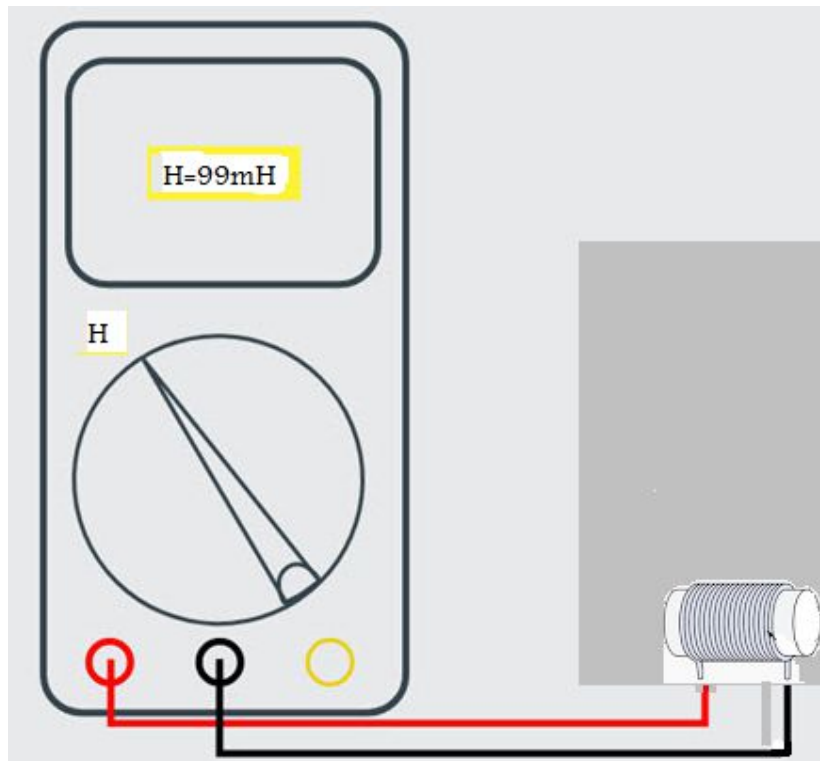
Capacitance measured in farad or microfarad . It is important when measuring capacitance that the circuit be de-energized or turned off, or the circuit may damage the meter. Most meters have overload protection on all ranges to prevent this, but you should check the specifications of your digital multimeter to be sure.



For capacitance measurements, place the test leads on each side of the capacitor. Other components in parallel with the capacitor being measured will have an effect on the measurement.

Inductance Measurements

Inductance measured in henry or milli henry. It is important when measuring Inductance that the circuit be de-energized or turned off, or the circuit may damage the meter. Most meters have overload protection on all ranges to prevent this, but you should check the specifications of your digital multimeter to be sure.

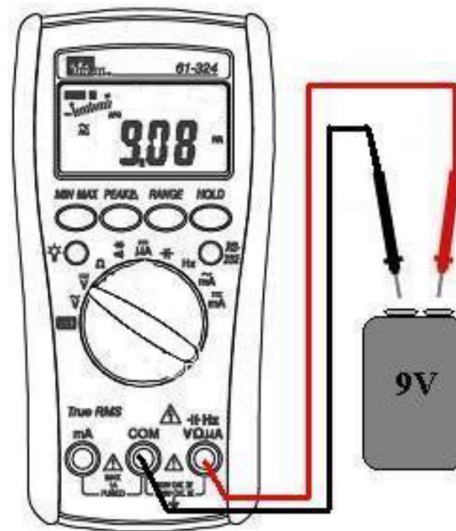


For inductance measurements, place the test leads on each side of the inductor.

DC Voltage Measurements:

To measure DC voltage, we place the Red lead into the **V Ω COM** port. Turn the dial or switch to V_{DC} or V_{-} .

We want to measure a 9V battery so the best range would be the 20 V range. If you have an auto-ranging meter you only need to set the function on the dial to V_{DC} or V_{-} .



Most Digital Multimeters are auto- polarity sensing devices. This means that we don't have to worry about having the Red lead on the hot or positive and the Black Lead on the Neutral or negative.

Measuring AC Voltage using Digital Multimeter:

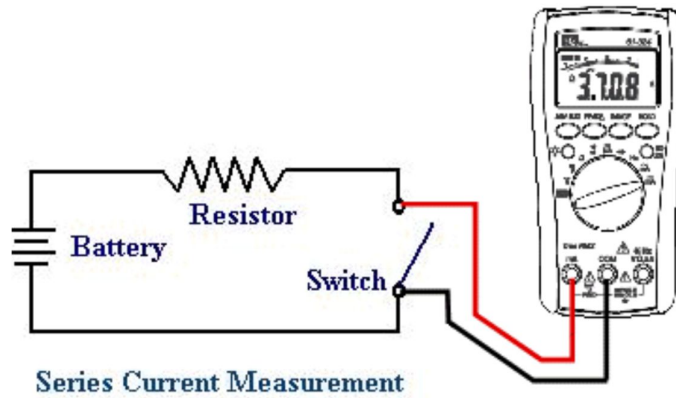
To measure the voltage, the Multimeter must be connected in parallel to the voltage source, load, or any circuit.

- Turn the meter ON by switching the ON/OFF button. Some turn ON by using the dial.
- Turn the dial to $\tilde{V}$ (V with a wavy symbol on top of it). It refers to AC voltage.
- If there is the availability of range selection (some meters have auto range mode for selecting range based on the reading), set the range to the largest expected value.
- The red probe must be inserted into its correct socket. Some meters have a dedicated socket for voltage while some have a combined socket for V-Amp- Ω . Insert the red probe into the socket having V on it.
- Place the black lead first on the lowest voltage point such as the ground.
- Place the red lead on the higher voltage point.
- Note the reading in the multimeter.
- If the range is set to the highest voltage range, reduce it step by step to achieve maximum resolutions and get an accurate reading.

Current Measurements :

When the equipment is turned on, it is considered to be a —load on the circuit. A load is any electrical component, such as a lamp, stereo, motor or heating element, that draws current. Current is measured in amperes, or amps.

To measure using a meter we must open the circuit and make the measurement in Series with the load..



To measure current with a multimeter, turn the power off at the breaker as close to the source as possible. Break the circuit, connect the multimeter in series with the circuit, and reestablish power.

Table: 1

a) Measured the values of resistances :

No of obs.	Theoretical values of Resistances (Ohms)	Measured values of resistances (Ohms)
1		
2		
3		
4		

Table: 2

a) Measured the values of Capacitance :

No of obs.	Theoretical values of Capacitance (Farad/ μ f)	Measured values of Capacitance (Farad/ μ f)
1		
2		
3		
4		

Table: 3

a) Measured the values of Inductance :

No of obs.	Theoretical values of Inductance (Henry/millinery)	Measured values of Inductance (Henry/millinery)
1		
2		
3		
4		

Table: 4

b) Measured the values of AC voltage:

No of obs.	Theoretical values of voltage (Volt)	Measured values of voltage (Volt)
1)		
2)		

Table: 5

c) Measured the values of DC voltage:

Measuring the voltage of a power supply:

No of obs.	Theoretical values of voltage (Volt)	Measured values of voltage (Volt)
1)		
2)		

Table: 6

d) Measured the values of current in a circuit :

No of obs.	Theoretical values of current (mA)	Measured values of current (mA)
1)		

Precautions

- 1) When you have to use multimeter first check proper DC polarity when measuring DC.
- 2) You have to always start with the highest voltage or current range.
- 3) Be sure that the multimeter is switched during the to AC prior to trying to measure AC circuits.
- 4) Set the ideal current range before measuring higher current or some else it will blow the digital multimeter fuse.
- 5) Never apply power to the circuit when you have to measure resistance with a multimeter.
- 6) In case you want to verify the presence of dangerous voltage in a circuit with a digital multimeter, it is crucial to verify both the AC as well as DC voltage.
- 7) Make sure that change the 0 ohms reading after changing resistance ranges before creating a resistance measurement.

Experiment (4)

To study a series LCR circuit and determine it's

a) Resonant frequency and b) Quality factor Q

circuits. Also to find the quality factor and band width in L-C-R series circuit.

Apparatus :- A non-inductive resistor (100 ohm), a capacitor (0.01 microfarad), inductor (100 milliHenry), a signal generator, an a.c. milli-ammeter and the connecting wires.

Theory :

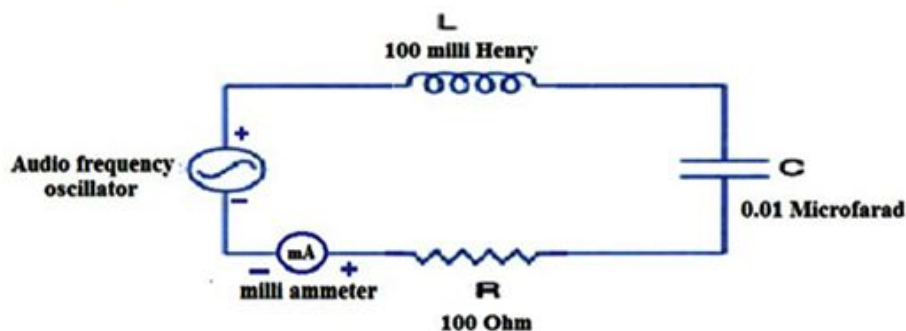


fig: Series LCR circuit diagram

The resonance frequency $f_0 = \frac{1}{2\pi\sqrt{LC}}$ Hz

Where L = Self inductance (mH)

C = Capacity of the capacitor (μF)

Quality factor $Q = \frac{2\pi f_0 L}{R}$

Where R = Resistance (Ω)

Band width = $(f_2 - f_1)$ (Hz)

Also Quality factor $Q = \frac{f_0}{f_2 - f_1} \Rightarrow Q = \frac{1}{R_t} \sqrt{\frac{L}{C}}$, $R_t = (R + \text{Resistance of inductance})$

Where f_1 and f_2 are the frequencies at the half power points.

(Series L-C-R) :

When the resistor R, inductor L and capacitor C are connected in series with a source of emf E, the circuit is called as the series resonant or series tuned circuit (figure-1). This is an acceptor circuit, that means it allows maximum current to flow through it at a particular (resonant) frequency and at all other frequencies it allows less current.

In A.C. circuits the voltage and the current are usually out of phase. Across the inductor, the current lags behind the voltage by 90° , where as across the capacitor, the current leads the voltage by 90° . But across the resistor the voltage and current both are in

phase. Under certain conditions, the voltage and current are in phase, even though the circuit consists of L , C and R and the circuit behaves as a pure resistor. This phenomenon is called resonance. This occurs at a single frequency known as resonant frequency. At this frequency the capacitive reactance ($X_c = 1/\omega C$) and the inductive reactance ($X_L = \omega L$) are equal and opposite in direction. So they get cancelled each other and only resistance acts.

The impedance of the circuit is given by $Z = R + j(\omega L - 1/\omega C)$

At resonance the reactive term disappears $\omega L - 1/\omega C = 0$

The impedance is minimum i.e. $Z = R$

The current is maximum $I = E/R$

$$\text{So } \omega L = 1/\omega C$$

$$\omega_0 = \omega = 2\pi f_0 = \frac{1}{\sqrt{LC}}$$

$$f_0 = \frac{1}{2\pi} \frac{1}{\sqrt{LC}}$$

At this frequency the current is maximum and this frequency f_0 is called resonant frequency. The circuit has selective properties. To compare selectivity or sharpness of resonance, a band of frequencies is chosen at which the current falls to $\frac{1}{\sqrt{2}}$ times (half power points) of its maximum value. The frequency difference ($f_2 - f_1$) between the half power points is called the bandwidth.

f_2 = Higher frequency

f_1 = Lower frequency

f_0 = Resonance frequency

Procedure :-

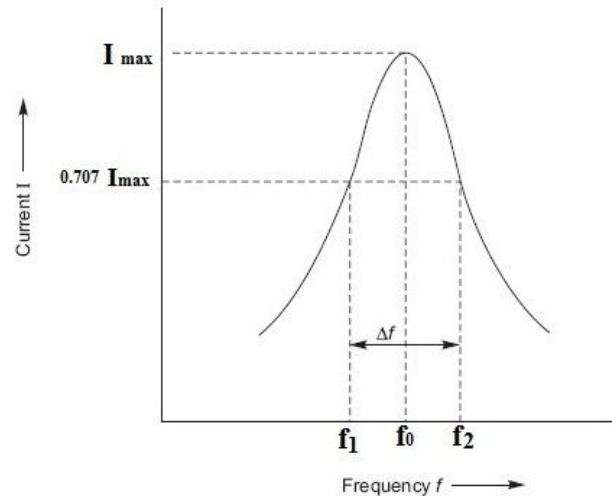
- 1) For L-C-R series, the circuit is connected as shown in the figure-1.
- 2) The source resistance and the series resistance should be small.
- 3) The out put voltage of the signal generator is adjusted to be around **1 Volt**
- 4) The frequency of the signal generator is changed in steps and the corresponding current values are noted from the a.c. milli- ammeter.
- 5) The current values increase with the increase of frequency, up to the resonant frequency further increase of frequency causes the decrease of current.
- 6) The L,C and R values are noted to calculate the resonant frequency f_0 and Q-factor, using the above formulae.

Table 1:

Frequency vs Current

Serial no	Frequency (Hz)	Current (mA)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		

Graph :- A graph is drawn for current against frequency. The frequency corresponding to maximum current is noted and it is the resonant frequency f_0 . The frequencies f_1 and f_2 corresponding to half power points is noted and from it the bandwidth, $(f_1 - f_2)$ is noted. From the values of f_0 , f_1 and f_2 , the quality factor, Q is calculated.



Calculation:

From the graph

Maximum current $I_{max} = \dots\dots\dots \text{mA}$

RMS value of current $I_{rms} = 0.707 \cdot I_{max} = \dots\dots\dots \text{mA}$

Resonance frequency $f_0 = \dots\dots\dots \text{Hz}$

Lower frequency $f_1 = \dots\dots\dots \text{Hz}$

Higher frequency $f_2 = \dots\dots\dots \text{Hz}$

Bandwidth:

$$(f_2 - f_1) = \dots\dots\dots \text{Hz}$$

Quality factor:

Experimental value of

$$Q = \frac{f_0}{f_1 - f_2}$$

$$= \dots\dots\dots$$

Again

Theoretical value of

$$Q = \frac{1}{R_t} \sqrt{\frac{L}{C}}$$

$R_t = (R + \text{Resistance of inductance})$

$= \dots\dots\dots$

Maximum Proportional Error:

$$Q = \frac{f_0}{f_1 - f_2}$$

$$\left(\frac{\delta Q}{Q}\right)_{max} = \frac{\delta f_0}{f_0} + \frac{\delta(f_1 - f_2)}{(f_1 - f_2)}$$

$$\left(\frac{\delta Q}{Q}\right)_{max} = \frac{\delta f_0}{f_0} + \frac{\delta(f)}{(f_1 - f_2)}$$

$\delta f_0 = 1$ small division of Audio frequency oscillator

$\delta(f) = 2 \times 1$ small division of audio frequency oscillator

Maximum % Error:

$$= \left(\frac{\delta Q}{Q}\right) \times 100$$

=.....%

Maximum Possible Error:

$$(\delta Q)_{max} = \left(\frac{\delta Q}{Q}\right)_{max} \times (\text{Measured value of } Q)$$

Results:

Resonant frequency of the series LCR circuit is $f_0 = \dots \text{Hz}$

Quality factor $Q = \dots$ (Theoretical)

$Q = [(\delta Q_{max}) \pm (\text{Measured value of } Q)]$ (Experimental, from the graph)

Precaution:

- 1) The internal resistance of the source and series resistance should be small
- 2) The voltage across C is Q times higher than input voltage. So it may be dangerously high when Q is large. So input voltage should be small
- 3) Increase the frequency in small steps to get correct value of I_{max} .

Experiment 5)

To study a parallel LCR circuit and determine its

a) Anti-resonant frequency and b) Quality factor Q.

Apparatus :- A non-inductive resistor (100 ohm), a capacitor (0.01 microfarad), res. inductor (100 milliHenry) a signal generator, an a.c. milli- ammeter and the connecting wi

Theory :

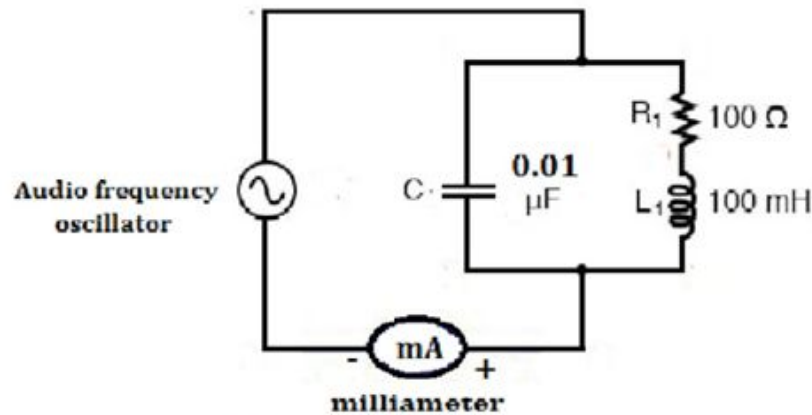


Fig : Parallel LCR circuit diagram

The resonance frequency $f_0 = \frac{1}{2\pi\sqrt{LC}}$ Hz

Where L = Self inductance (mH)

C = Capacity of the capacitor (μF)

Quality factor $Q = \frac{1}{R_t} \sqrt{\frac{L}{C}}$,

Where $R_t = (R + \text{Resistance of inductance})$

Band width = $(f_2 - f_1)$ (Hz)

Also Quality factor $Q = \frac{f_0}{f_2 - f_1}$

Where f_1 and f_2 are the frequencies at the half power points.

L-C-R parallel :- Parallel resonant circuit (figure-2) is one in which one branch consists of an inductor L with associated resistor R and the other branch consists of a capacitor C. This is a rejector circuit, that means it rejects the current or allows minimum current to flow through it, at a particular (anti- resonant) frequency and it allows more current at all other frequencies. So the circuit is not selective. But it is highly selective when energized from a high impedance generator.

The impedance of the circuit is given by

$$\frac{1}{Z} = \frac{1}{R + j\omega L} + \frac{1}{1/j\omega C}$$

At resonance the impedance is maximum.

The impedance at resonance

$$Z = \frac{L}{CR}$$

The anti-resonance frequency

$$f_o = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}} \text{ Hz}$$

If R-value is small, then

$$f_0 = \frac{1}{2\pi \sqrt{LC}} \text{ Hz}$$

At this frequency the current is minimum and this frequency f_0 is called anti-resonant frequency. The circuit has selective properties. To compare selectivity or sharpness of resonance, a band of frequencies is chosen at which the current rise to $\sqrt{2}$ times of its minimum value. The frequency difference ($f_2 - f_1$) between the half power points is called the bandwidth

$$\text{Quality factor: } Q = \frac{R}{2\pi f L} = 2\pi f C R = \left(\frac{1}{R_t}\right) \sqrt{\frac{C}{L}}$$

Procedure:

- 1) the circuit is connected as shown in the figure
- 2) The frequency of the signal generator is changed in steps and the corresponding current values are noted from the a.c. milli- ammeter.
- 3) The readings are tabulated
- 4) The current values decrease with the increase of frequency up to the anti-resonant frequency
- 5) Further increase of frequency causes the increase of current.
- 6) The anti- resonant frequency f_0 is noted corresponding to the minimum current in the circuit.

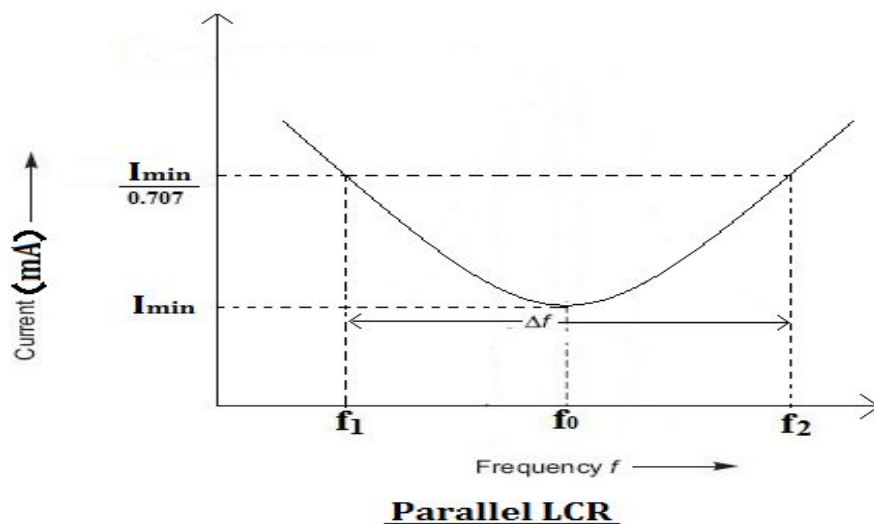
Table:

Frequency vs. Current

[illegible]

Plotting of graph:

A graph is drawn for current against frequency. The frequency corresponding to minimum current is noted and it is the anti-resonant frequency f_0 . The frequencies f_1 and f_2 corresponding to $0.707 \cdot I_{\min}$ noted and from it the bandwidth, $(f_1 - f_2)$ is noted. From the values of f_0 , f_1 and f_2 , the quality factor, Q is calculated.



Calculation:

From the graph

Maximum current $I_{\min} = \dots\dots\dots \text{mA}$

RMS value of current $I_{\text{rms}} = I_{\min} / 0.707 = \dots\dots\dots \text{mA}$

Anti Resonance frequency $f_0 = \dots\dots\dots \text{Hz}$

Lower frequency $f_1 = \dots\dots\dots \text{Hz}$

Higher frequency $f_2 = \dots\dots\dots \text{Hz}$

Bandwidth:

$$(f_2 - f_1) = \dots\dots\dots \text{Hz}$$

Quality factor:

Experimental value of $Q = \frac{f_0}{f_1 - f_2}$
=.....

Again

Theoretical value of

$$Q = \frac{1}{R_t} \sqrt{\frac{C}{L}}$$

$R_t = (R + \text{Resistance of inductance})$
=.....

Maximum Proportional Error:

$$Q = \frac{f_0}{f_1 - f_2}$$
$$\left(\frac{\delta Q}{Q}\right)_{\max} = \frac{\delta f_0}{f_0} + \frac{\delta(f_1 - f_2)}{(f_1 - f_2)}$$

$$\left(\frac{\delta Q}{Q}\right)_{max} = \frac{\delta f_0}{f_0} + \frac{\delta(f)}{(f_1 - f_2)}$$

$\delta f_0 = 1$ small division of Audio frequency oscillator

$\delta(f) = 2 \times 1$ small division of audio frequency oscillator

Maximum % Error:

$$= \left(\frac{\delta Q}{Q}\right) \times 100$$

$$= \dots\dots\dots\%$$

Maximum Possible Error:

$$(\delta Q)_{max} = \left(\frac{\delta Q}{Q}\right)_{max} \times (\text{Measured value of } Q)$$

Results:

Resonant frequency of the series LCR circuit is $f_0 = \dots\dots\text{Hz}$

Quality factor $Q = \dots\dots\dots$ (Theoretical)

$Q = [(\delta Q_{max}) \pm (\text{Measured value of } Q)]$ (Experimental, from the graph)

Precaution:

- 1) The internal resistance of the source and series resistance should be small
- 2) The voltage across C is Q times higher than input voltage. So it may be dangerously high when Q is large. So input voltage should be small
- 3) Increase the frequency in small steps to get correct value of I_{min} .

Experiment (7)

Determination of Powers of Given lenses using an Optical Bench

- (i) Convex Lens
- (ii) Concave Lens

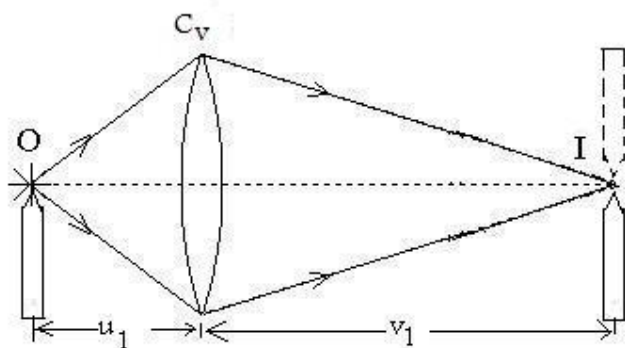


Fig1. Ray diagram to determine the focal length and power of a convex lens

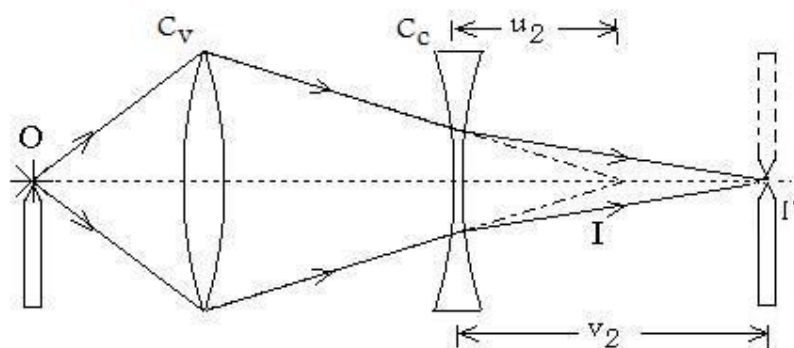


Fig2. Ray diagram to determine the focal length and power of a concave lens

Theory: If an object (O) is placed at a distance (greater than the focal length of the lens) from the convex lens (C_v) real inverted image (I) of the object is formed on the other side of the lens. Then the focal length of the convex lens can be determined by the relation.

$$f = \frac{uv}{u - v} \dots\dots\dots(1)$$

Where, $u = u_1$ = object distance

$v = v_1$ = image distance for the lens.

Then Power of the lens is

$$P = 100/f \text{ dioptre} \dots\dots\dots(2)$$

Now, if a concave lens (C_c) is placed in between the lens C_v and image (I) in such a position that the image (I) is within the focal length of the concave lens (C_c), then real image (I) will act as a virtual source to the concave lens, producing its real image (I'). Using the same formulae as given above, the focal length and power of the concave lens can be determined.

For Concave lens... $u = u_2$ = object distance

$v = v_2$ = image distance for the lens.

Apparatus Used:

(i) Optical bench, (ii) two lens holders and two pin holders with pins, (iii) one concave and one convex lens, (iv) index rod etc.

Procedure:

- (i) The optical bench is arranged in such a way that the tips of the two pins and the optical centre of the two lenses lie at the same height. All the four points should be in a straight line parallel to the optical bench.
- (ii) The length of the index rod is measured. Then the apparent length of the rod is measured for both the lenses by placing the rod in between object pin and the centre of the convex lens and concave lens separately. Index error for both the lens can be determined.
- (iii) The convex lens C_v is mounted on the central stand and placed nearly at the middle of the optical bench. Two pins O (object) and I (image) are put on two stands one on either side of the lens.
- (iv) The object distance is changed in such a way that an inverted real image is formed on the other side of the lens. The position of the image pin is changed till parallax is removed. The position of the image pin is noted from the bench. Using the readings of object distance u_1 , image distance v_1 ; focal length f_1 is calculated.
- (v) The object distance is maintained at $2f_1$ and the image position is again determined as earlier step. The position of the image is noted.
- (vi)
- (vii) Then the concave lens C_c is placed between the convex lens C_v and the image I. Placing the eye in the line with the tip of the pin the image is observed and the position of image pin is adjusted till there is no parallax
- (viii) Position of the image is noted.

Determination of Index Error:

Distance from the bench (L_1):cm (with the rod between object and centre of convex lens)

Distance from the bench (L_2):cm (with the rod between object and centre of concave lens)

Index error convex lens: $x_1 = L_1 - L = \dots$ cm

Index error concave lens: $x_2 = L_2 - L = \dots$ cm

Table 1: Reading for focal length of the convex lens-

[illegible][illegible]

CALCULATION :

Power of the convex lens $P_1 = \frac{100}{f_1}$ dioptre

Power of the convex lens $P_2 = \frac{100}{f_2}$ dioptre

ERROR CALCULATION:**Maximum Proportional Error:**

We have, $f = uv/(u-v)$

$$P = \frac{1}{f} = \frac{u-v}{uv}$$

$$(\delta P / P)_{\max} = \delta u / u + \delta v / v + \delta(u-v) / (u-v)$$

$$= \delta u / u + \delta v / v + (\delta u + \delta v) / (u-v) \text{ -----(1)}$$

Here,

$\delta u = \delta v = 2 \times$ least graduation on the scale of the optical bench.

Using Equation (1) determine proportional error for both the lenses.

Maximum Percentage Error = Maximum Proportional Error $\times 100 \%$

Maximum Possible Error for convex lens (δP_1) = $(\delta P_1 / P_1)_{\max} \times$ measured value of P_1 dioptre.

Maximum Possible Error for concave lens (δP_2) = $(\delta P_2 / P_2)_{\max} \times$ measured value of P_2 dioptre.

RESULT:

The measured value of the power of the given convex lens is found to be

= (Measured value of $P_1 \pm \delta P_1$) dioptre.

The measured value of the power of the given concave lens is found to be

= (Measured value of $P_2 \pm \delta P_2$) dioptre.

PRECAUTIONS:

- (i) While determining the position of the image, parallax should be avoided.
- (ii) Both the lenses should be thin and of small aperture having small focal length.
- (iii) Index error is same for the object and the image distance.

Expt. No (8)

To determine the radius of curvature of the curved surface of the plano convex lens or the wavelength of the source of light by Newton's Ring method.

Apparatus Used: Newton's ring apparatus, monochromatic light source

Theory:

When a parallel beam of monochromatic light is incident normally on a combination of a plano-convex lens L and a glass plate G, as shown in Fig.(a), a part of each incident ray is reflected from the lower surface of the lens, and a part, after refraction through the air film between the lens and the plate, is reflected back from the plate surface. These two reflected rays are coherent, hence they will interfere and produce a system of alternate dark and bright rings with the point of contact between the lens and the plate as the center. These rings are known as Newton's ring.

If D_n and $D_{(n+m)}$ be the diameters of n^{th} and $(n + m)^{\text{th}}$ bright or dark rings respectively, then the radius of curvature of the curved surface of the plano convex lens is defined as-

$$R = \frac{(D_{n+m})^2 - (D_n)^2}{4\lambda m} \dots\dots\dots(1)$$

Where λ is the wavelength of the monochromatic light used.

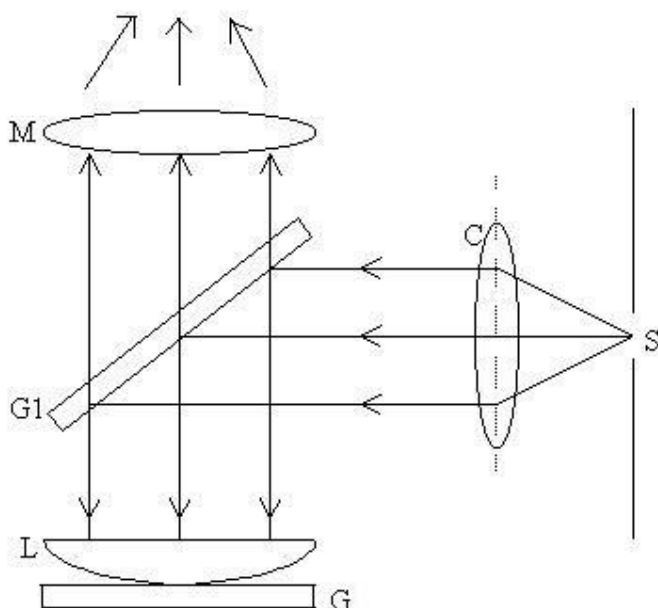
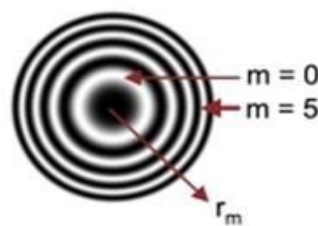


Fig (a). Experimental setup to observe Newton's ring



Fig(b). Newton's ring

Procedure :

1. At first preliminary adjustments are done as shown in fig (a), to observe the Newton's rings properly.
2. Now we adjust the glass plate G1 for maximum visibility of the point of contact of lens L with the glass plate G and hence for maximum visibility of Newton's Rings. In this orientation, G1 is at 45° to the incident beam of light.

[illegible]

Calculation:

We have
$$R = \frac{(D_{n+m})^2 - (D_n)^2}{4\lambda m} \quad \text{cm}$$

Error Calculation:**Maximum Proportional Error:**

$$\left(\frac{\partial R}{R} \right)_{\max} = 2 \times \frac{\partial D_{n+m}}{D_{n+m}} + 2 \times \frac{\partial D_n}{D_n}$$

$$\text{Maximum \% Error} = \left(\frac{\partial R}{R} \right)_{\max} \times 100\%$$

$$\text{Maximum Possible Error, } \partial R = \left(\frac{\partial R}{R} \right)_{\max} \times \text{measured value of 'R'}$$

Result: The measured value of radius of curvature of the curved surface of the plano-convex lens is found to be = $(R \pm \partial R)$ cm

Precautions:

1. The set up must not be disturbed in any way during experiment.
2. The microscope must always be moved in one direction in a particular set of taking the readings, so as to avoid any back-lash error.
3. A few rings near the centre of the pattern should be avoided while taking the readings.

Experiment (9)

To find the refractive index of water by using a convex lens and a mirror.

Apparatus Used:

Convex lens (focal length nearly 20 cm), plane mirror (bigger in size than the aperture of the lens), laboratory stand fixed with a pin in clamp, metre scale.

Theory:

If a convex lens is placed over a few drops of liquid placed on the surface of a plane mirror, then a plano-concave liquid lens is formed between the lower surface of the convex lens and the plane mirror. The combination behaves as a converging lens since the focal length of the plano-concave lens is greater than the focal length of the convex lens. If f_1 and f_2 be the focal length of glass convex lens and liquid lens and F be the focal length of their combination then,

$$\begin{aligned}\frac{1}{F} &= \frac{1}{f_1} + \frac{1}{f_2} \\ \Rightarrow \frac{1}{f_2} &= \frac{1}{F} - \frac{1}{f_1} = \frac{f_1 - F}{Ff_1} \\ \Rightarrow f_2 &= \frac{Ff_1}{f_1 - F} \dots\dots\dots(1)\end{aligned}$$

According to lens maker's formula,

$$\frac{1}{f_2} = (\mu - 1) \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \dots\dots\dots(2)$$

Where, μ is the refractive index of the plano- concave lens (liquid).

Here,

$r_2 = \infty$, (for the plane surface of the plano- concave lens)

let $r_1 = r$ (radius of the concave surface of the plano- concave lens)

eqn. (2) becomes....

$$\begin{aligned}\frac{1}{f_2} &= (\mu - 1) \left(\frac{1}{r} \right) \\ \Rightarrow \mu &= 1 + \frac{r}{f_2} \dots\dots\dots(3)\end{aligned}$$

Hence, by finding the focal length (f_2) of the plano-concave lens by coincidence method and radius of curvature (r) of its any one surface, the value of μ of the liquid can be found out from eqn.(3).

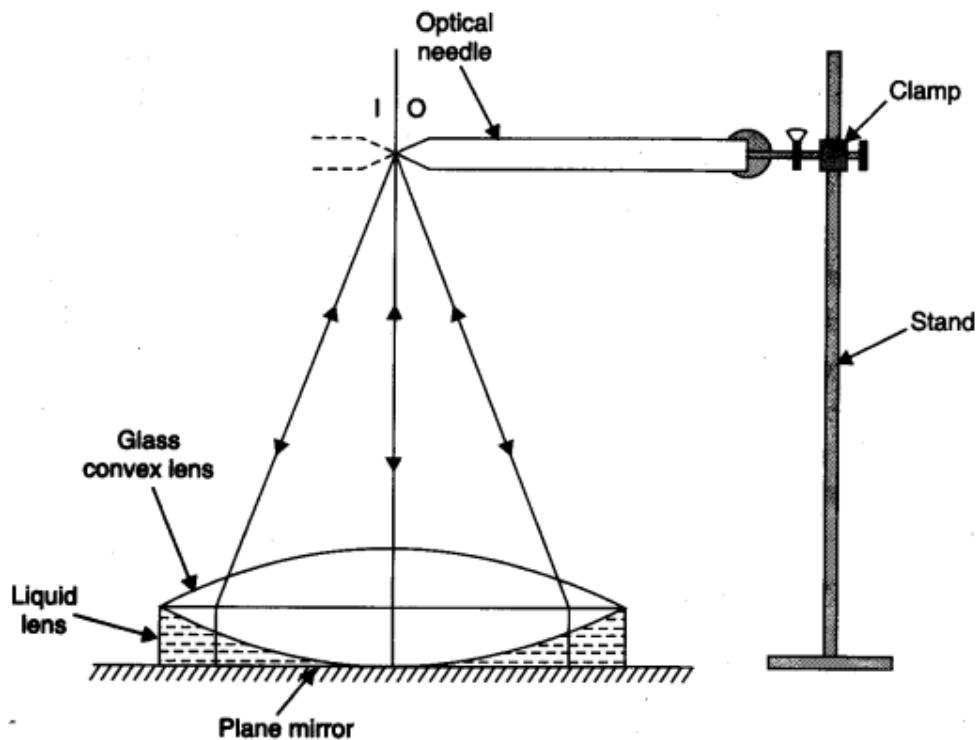


Fig. Focal length of glass convex lens and liquid lens combination.

Procedure:

1. The thickness of the lens at its centre is measured by a slide callipers. Half of the thickness (t) is to be added to the obtained focal length of the lens by coincidence method, to get the correct focal length from the optical centre.
2. The lens is placed on the plane mirror and its focal length (f_1) is determined by the coincidence method (for a definite position of object pin, the incident ray after refraction through the lens, is incident normally on the mirror. So, the reflected ray retrace the path and as a result real image is formed without parallax by touching the tip of the needle to the tip of the image) using a pin.
3. A few drops of experimental liquid are placed on the plane mirror and the convex lens is put over it as shown in figure. (A plano concave liquid lens is formed between plane mirror and convex lens).
4. Each time, the value of $\frac{t}{2}$ is added to the value of ' f_1 ' and ' F ' to obtain their correct values.
5. Step (2) to (4) are repeated at least three times to by displacing the pin position.

Observation :

Table (1): Determination of the thickness of the lens.

$$V.C. = \frac{1}{n} \times 1MSD = \dots\dots\dots\text{cm.}$$

$$I.E. (X) = \pm x \times V.C. \quad \text{cm} \quad (\text{where})$$

x= The VSD that coincide with any MSD when the jaws are closed

n= number of divisions of the venire scale)

No. of obs.	M.S.R. (A) cm	V.S.D. (B)	V.S.R. C = B×V.C. cm	Total thickness T = A+C cm	I.E. ±X cm	Corrected thickness t = T – (±X) cm	Mean t cm.	$\frac{t}{2}$ cm.
1								
2								
3								
4								

Table (2): Determination of the focal length (F) of the combined lens:

Sl. No.	when the pin and its real image coincide						Focal length of the plano-concave lens $f_2 = \frac{Ff_1}{f_1 - F}$ cm
	For convex lens			For combined lens			
	Distance of pin from upper surface of the lens (d ₁) cm	Mean Distance (d ₁) cm	Focal length (f ₁ =d ₁ + $\frac{t}{2}$) cm	Distance of pin from upper surface of the lens (d ₂) cm	Mean Distance (d ₂) cm	Focal length (F=d ₂ + $\frac{t}{2}$) cm	
1							
2							
3							

Calculation:

$$\text{The value of refractive index of the liquid is } \Rightarrow \mu = 1 + \frac{r}{f_2}$$

ERROR CALCULATION:

Maximum Proportional Error:

$$\text{We have, } \mu = 1 + \frac{r}{f_2}$$

$$\begin{aligned}\left(\frac{\partial \mu}{\mu}\right)_{\max} &= \frac{\partial f_2}{f_2} = 2 \times \left(\frac{\partial F}{F} + \frac{\partial f_1}{f_1} \right) \\ &= 2 \times \left[\frac{\partial d_1}{d_1} + \frac{\partial d_2}{d_2} + 2 \times \frac{\partial t}{t} \right]\end{aligned}$$

Maximum Percentage Error: $\left(\frac{\partial \mu}{\mu}\right)_{\max} \times 100\%$

Maximum Possible Error: $(\partial \mu)_{\max} = \left(\frac{\partial \mu}{\mu}\right)_{\max} \times \text{Calculated value of } \mu$

Result : The measured value of refractive index of the liquid is found to be $= (\mu \pm \partial \mu)$

Precautions:

1. Parallax error must be avoided while taking readings for determination of focal length.
2. The tip of the pin must lie along the axis of the lens.
3. Real image (inverted) must be formed by the lens and the mirror.

Experiment (10)

To find the refractive index of the material of a prism using a spectrometer (by finding the angle of the prism and the angle of minimum deviation of the prism).

Theory:

If a ray of monochromatic light while passing through a prism, has an angle of minimum deviation δ_m and if A be angle of the prism, then the refractive index (μ) of the material of the prism is given by

$$\mu = \frac{\sin \{(A + \delta_m)/2\}}{\sin A/2}$$

Hence, measuring the angle of minimum deviation (δ_m) of the ray passing through the prism and the refracting angle (A) of the prism, the value of μ can be determined.

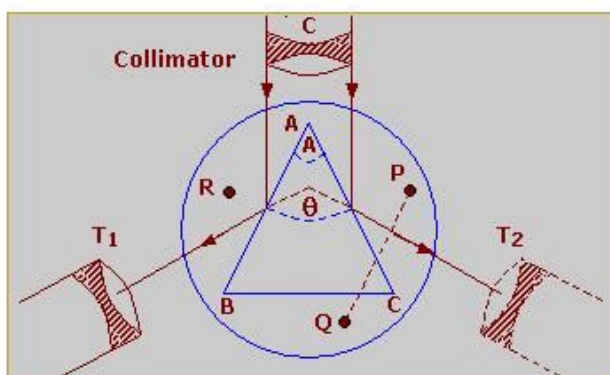


Fig (a): Determination of Angle of Prism

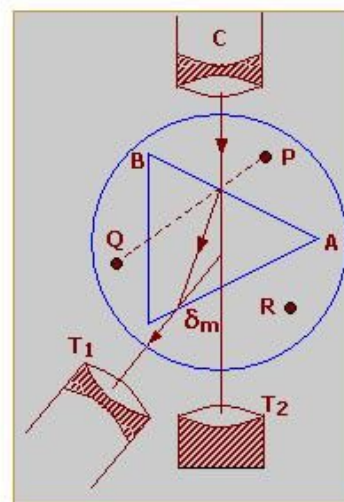


Fig (b): Determination of angle of minimum deviation

Apparatus used:

(i) Spectrometer (ii) prism (iii) spirit level (iv) source of light (v) magnifying glass

Procedure:

We can divide the entire procedure of this experiment in to the following two parts:

A) Determination of the angle of prism (A)

1. We have to Place the prism on the prism table with its refracting angle A towards the collimator. In this case some of the light falling on each face will be reflected and can be received with the help of the telescope.
2. The telescope is moved to one side to receive the light reflected from the face AB and the cross wires are focused on the image of the slit. The readings of the two scales are taken.
3. The telescope is moved to the other side to receive the light reflected from the face AC and again the

cross wires are focused on the image of the slit. Again the readings are noted.

4. The angle through which the telescope is moved; or the difference in the two positions gives twice of the refracting angle A of the prism. Therefore half of this angle gives the refracting angle of the prism.

B) Determination of the angle of minimum deviation (δ_m) of the prism.

1. The prism is placed in such a way that one of its faces (AB) points towards the collimator and we get the refracted image. If the prism table is gradually rotated and the slit-image is viewed through the telescope, it will be found that the image is deviated up to a certain point and then comes back in the same direction. The point from where the image returns back, is the position of minimum deviation. The cross-wire is made to coincide with the image at this point and the readings in both the scales are taken. Say, the mean of the readings is X_2
2. The prism is then removed and the telescope is focused to get the direct image and again the readings in both the scales are taken. Say, the mean of the readings is X_1 . The difference of X_1 and X_2 gives the angle of minimum deviation.
- 3.

Observation:

1) Determination of vernier constant:

Magnitude of smallest circular scale division = 1MSD=-----°
 No. of Vernier divisions N = -----
 $V.C = 1/N \times 1MSD =$ -----°

2) Table: Determination of angle of prism:

Scale	Reading for								Difference of reading 2A= (R ₁ ~ R ₂) (degree)	Mean 2A (degree)	Angle of prism A (degree)
	First image at T ₁				Second image at T ₂						
	Main Scale (a) deg	V.S.D. (b)	Value of VSR= b x V.C. deg	Total (R ₁) deg	Main Scale (a) deg	V.S.D. (b)	Value Of VSR= b x VC deg	Total (R ₂)			
First	...	...	...	...	...	...	...	...	...		
	...	...	...	...	...	...	...	...	...		
	...	...	...	...	...	...	...	...	...		
Second	...	...	...	...	...	...	...	...	...		
	...	...	...	...	...	...	...	...	...		
	...	...	...	...	...	...	...	...	...		

3) Table: Determination of the angle of minimum deviation (δ_m):

Scale	Reading for								Difference Of reading ($X_1 \sim X_2$) i.e δ_m (degree)	Mean δ_m in degree
	Direct Image				Deviated Image					
	Main Scale (a) deg	V.S.D. (b)	Value of VSR= b x V.C. deg	Total (X_1) deg	Main Scale (a) deg	V.S.D. (b)	Value Of VSR= b x VC deg	Total (X_2)		
First	...	...	...	...	...	...	...	...	...	
	...	...	...	...	...	...	...	...	...	
	...	...	...	...	...	...	...	...	...	
Second	...	...	...	...	...	...	...	...	...	
	...	...	...	...	...	...	...	...	...	
	...	...	...	...	...	...	...	...	...	

Calculation:

Refractive index $\mu = \sin \{(A + \delta_m)/2\} / \sin A/2$

=

Error Calculation:

Maximum Proportional Error :

$$(\delta \mu / \mu)_{\max}$$

We have, $\mu = \sin \{(A + \delta_m)/2\} / \sin A/2$

$$(\delta \mu / \mu)_{\max} = \delta \{ \sin (A + \delta_m)/2 \} / \{ \sin (A + \delta_m)/2 \} + \delta \{ \sin A/2 \} / \sin A/2$$

$$= \cot(A + \delta_m)/2 \cdot \delta(A + \delta_m)/2 + \cot A/2 \cdot \delta (A/2)$$

$$= \left[\cot \left(\frac{A + \delta_m}{2} \right) \cdot \left\{ \partial \left(\frac{A}{2} \right) + \partial \left(\frac{\delta_m}{2} \right) \right\} + \cot \frac{A}{2} \cdot \partial \left(\frac{A}{2} \right) \right] \times \frac{3.14}{180}$$

=

Here, $\delta (\delta_m/2) = \delta (A/2) = (2 \times V.C.)$

Maximum percentage error= $(\delta \mu / \mu)_{\max} \times 100 \%$

Maximum possible error $(\delta \mu) = (\delta \mu / \mu)_{\max} \times$ Measured value of μ

Result: The measured value of μ is found to be $= (\mu \pm \delta \mu) = \dots\dots\dots$

Precautions:

1. While determining angle of prism (A) the vertex of prism should be placed slightly ahead of the centre

of the prism table towards the collimator.

2. During rotation of the telescope, if the zero of the vernier scale crosses the zero mark of the main circular scale, then the latter should be considered as 360° and calculations should be done accordingly.
3. There should not be any parallax between the cross-wire and the slit image.